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MATHEMATICS

TARGET : JEE (Main) 2024

CHAMP-2024 REVISION DPP NO. 01

Syllabus : FOM & Trigonometry

SECTION - I

Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

1. The least integral value of 'a' such that $(a - 2) x^2 + 8x + a + 4 > 0 \ \forall \ x \in R$ is
(1) 3 (2) 5 (3) 4 (4) 6

2. The number of real solutions of equation $2^{x/2} + \left(\sqrt{2}+1\right)^x = \left(5+2\sqrt{2}\right)^{x/2}$ is
(1) 1 (2) 2 (3) 4 (4) infinite

3. If $y \neq 0$, then the number of values of the pair (x, y) such that $x + y + \frac{x}{y} = \frac{1}{2}$ and $(x+y)\frac{x}{y} = -\frac{1}{2}$, is
(1) 1 (2) 2 (3) 0 (4) 3

4. If the number of consecutive odd integers whose sum can be expressed as $50^2 - 13^2$ is k then k can be
(1) 33 (2) 35 (3) 37 (4) 39

5. If α, β are the roots of the quadratic equation $x^2 - \left(3 + 2^{\sqrt{\log_2 3}} - 3^{\sqrt{\log_3 2}}\right)x - 2\left(3^{\log_3 2} - 2^{\log_2 3}\right) = 0$, then the value of $\alpha^2 + \alpha\beta + \beta^2$ is equal to
(1) 11 (2) 7 (3) 3 (4) 5

6. The factors of $\sin\theta + \sin\phi - \cos\theta \sin(\theta + \phi)$ are
(1) $\sin\theta$ and $1+\sin(\theta+\phi)$
(2) $\sin\theta$ and $1 - \cos(\theta + \phi)$
(3) $\sin\theta$ and $1 + \cos(\theta+\phi)$
(4) None of the above

7. The coefficient of x in polynomial $(X - \alpha)(X - \beta)(X - \gamma)$ where $\alpha = \cos 75^o$, $\beta = \cos 45^o$ and $\gamma = \cos 165^o$
(1) $\frac{2\sqrt{2}+1}{4}$ (2) $\frac{-2\sqrt{2}+1}{4}$ (3) $\frac{3}{4}$ (4) $\frac{-3}{4}$

8. Which one of the following is correct?
(1) $6\cos 20° - 8\cos^3 20° = 1$
(2*) $8\sin^3 10° - 6\sin 10° = -1$
(3) $6\cos^3 20° - 8\cos 20° = 1$
(4) $8\sin 10° - 6\sin^3 30° = 1$

9. If $\cos(\alpha+\beta) = 0$ then $\text{cosec}(\alpha-\beta) =$
(1) $\cos 2\beta$ (2) $\sec 2\beta$ (3) $-\sec 2\beta$ (4) $\pm \sec 2\beta$

10. If $\tan\alpha = \frac{1+\sqrt{1+\sin 2\theta}}{1-\sqrt{1-\sin 2\theta}}$ where $0 < \theta < \frac{\pi}{4}$ then value of α can be
(1) $\frac{\pi}{2} - \frac{\theta}{2}$ (2) $\frac{\pi}{4} + \frac{\theta}{2}$ (3) $\pi + \theta$ (4) θ

11. If $a \leq 3\cos x + 5\sin\left(x - \frac{\pi}{6}\right) \leq b$ holds good for all x where length of interval [a,b] is minimum then a + 2b =
(1) $\sqrt{19}$ (2) $\sqrt{34}$ (3) 4 (4) 8

12. If 2 cos θ + sin θ = 1 $\left(\theta \neq \frac{\pi}{2}\right)$ then 7 cos θ + 6 sin θ is equal to :

(1) $\frac{1}{2}$ (2) 2 (3) $\frac{11}{2}$ (4) $\frac{46}{5}$

13. If sin 75° – cos 105° $\neq$ cos θ then

(1) $\theta \in (0, 60^o)$ (2) $\theta \in (0, \frac{\pi}{2})$ (3) $\theta \in I$ (4) $\theta \in R$

14. If E = sin θ + cos θ then

(1) $E < 1$ for $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ (2) $E > \sqrt{2}$ for $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

(3) $1 < E > \sqrt{2}$ for $\theta \in \left(0, \frac{\pi}{4}\right)$ (4) For $\theta \in \left(0, \frac{\pi}{4}\right)$, range of E is $\left(0, \sqrt{2}\right)$

15. If α, β are acute angles and tan(α + β) = 1 and sec (α – β) = 2 then tan 2α = (where α < β)

(1) $\cot\frac{7\pi}{12}$ (2) $\tan\frac{\pi}{12}$ (3) $\cot\frac{\pi}{12}$ (4) $-\cot\frac{\pi}{12}$

16. **Statement-1 :** The maximum value of

$\frac{1}{3\cos x + 4\sin x + 1}$ is $\frac{1}{6}$

Statement-2 :

$-\sqrt{a^2 + b^2} \leq a \sin x + b \cos x \leq \sqrt{a^2 + b^2}$.

(1) Statement -1 is True, Statement -2 is True ; Statement -2 is a correct explanation for Statement -1
(2) Statement-1 is True, Statement-2 is True ; Statement-2 **is NOT** a correct explanation for Statement-1
(3) Statement -1 is True, Statement -2 is False
(4) Statement -1 is False, Statement -2 is True

17. If f(θ) = $\cos^2 \theta + \sin^4 \theta$. Then minimum value of f(θ) is

(1) $\frac{1}{4}$ (2) $\frac{3}{4}$ (3) 1 (4) $\frac{1}{2}$

18. If cos α + cos β = $\frac{3}{2}$ and sin α + sin β = $\frac{1}{2}$ and θ is the arithmetic mean of α and β, then sin 2θ + cos 2θ is equal to :

(1) $\frac{3}{5}$ (2) $\frac{7}{5}$ (3) $\frac{4}{5}$ (4) $\frac{8}{5}$

19. The value of $\frac{1}{\cos 285^o} + \frac{1}{\sqrt{3}\sin 255^o}$ is

(1) $\sqrt{3} - \sqrt{2}$ (2) $2\sqrt{2}$ (3) $\frac{4\sqrt{2}}{\sqrt{3}}$ (4) $\frac{2\sqrt{2}}{3}$

20. Let P denotes the value of cosα – cos2α + cos3α where α = $\frac{\pi}{7}$ and Q denotes the value of 8cosα cos2α cos4α where α = $\frac{\pi}{7}$ then $P^2 + Q^2$ equals

(1) $\frac{1}{4}$ (2) $\frac{3}{4}$ (3) $\frac{5}{4}$ (4) $\frac{7}{4}$

SECTION-II
(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. Find the number of real solutions of equation $\sqrt{x^2-4x+3}+\sqrt{x^2-9}=\sqrt{4x^2-14x+6}$

22. The number of positive integers satisfying the equation $x + \log_{10}(2^x + 1) = x\log_{10}5 + \log_{10}6$ is-

23. For how many distinct values of A between 0° and 360° is the expression $\frac{\sin A+\sin 2A+\sin 3A}{\cos A+\cos 2A+\cos 3A}$ undefined?

24. If $3\tan(\theta - 30^\circ) = \tan(\theta + 120^\circ)$, then value of $\cos 2\theta$ equals

25. If $\cos A = \frac{3}{4}$, then the value of $16\cos^2\left(\frac{A}{2}\right) - 32\sin\left(\frac{A}{2}\right)\sin\left(\frac{5A}{2}\right)$ is :

ANSWER KEY_DPP-01

1.	**(2)**	**2.**	**(1)**	**3.**	**(2)**	**4.**	**(3)**	**5.**	**(2)**	**6.**	**(2)**	**7.**	**(4)**
8.	**(2)**	**9.**	**(4)**	**10.**	**(1)**	**11.**	**(1)**	**12.**	**(2)**	**13.**	**(4)**	**14.**	**(3)**
15.	**(1)**	**16.**	**(4)**	**17.**	**(2)**	**18.**	**(2)**	**19.**	**(3)**	**20.**	**(3)**	**21.**	**(1)**
22.	**(1)**	**23.**	**(6)**	**24.**	**(1)**	**25.**	**(3)**						